The Minimal Generating Set — one page

Two invariants generate the Standard Model's discrete skeleton. One measured number picks the world. *(#66 · 2026-08-24 · from banked v0.7/K1697; no new claims)*

The plain version first

Ask what you must be told about BST's geometry to reconstruct the discrete Standard Model — not the constants, the *skeleton*: which forces, which charges, why left-handed, why no sterile neutrino. The answer, proved by census and refutation rather than assembled by hand, is: two facts about the shape, plus one number from experiment.

The two invariants

	invariant	what it generates
I_1	the census — which division algebras the domain's endomorphisms realize: End_K = $\mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}$	the three-force structure; the rank (= 2) is DERIVED from it, not assumed
I_2	non-orientability of the Shilov boundary (n odd; Pin, not Spin)	chirality — the one-sided edge that makes a left-handed world possible

Minimality is a theorem, not a preference (v0.7 Appendix A): the generating set has size exactly two — SEP-3 (a proposed third independent generator) is refuted, and rank is a consequence of I_1, collapsing what looks like three inputs into two.

Where to check this claim inside this packet: the companion Forcing & Evidence paper, Part I ("D_IV^5 is the minimal generating set for a physics-capable reality"), carries the two-leg forcing argument and its per-leg tiers — Leg A the family selection, Leg B the census selection within rank 2 (N_c = rank^2 − 1, physics-free). In brief: the census I_1 determines the rank (so "rank" is not a third generator — it is a consequence), and the candidate third invariant SEP-3 fails because a separator must distinguish members of the family and SEP-3 provably cannot — a refutation, not a preference. The full proof is the source paper's Appendix A; the packet-level support is F&E Part I.

The three conditions and the survivors

The two invariants impose three proved conditions on the dimension n: quaternionic spinor ($n \equiv 3, 4, 5 \bmod 8$ — gives the weak SU(2)) · odd n (one-sided boundary — gives chirality) · colour exists ($n - 2 > 1$). Survivors: $n \in \{5, 11, 13, 19, 21, \ldots\}$.

The ruler

One measured input — $N_c = 3$ colours — pins $n = 5$ (colour count $= n - 2$). This is the program's single declared input, and it is stated as an input everywhere, never smuggled: *the geometry narrows the world to a short list; one measurement reads which line of the list we live on.*

What this page does NOT claim

Not the mass values, not the mixing angles, not the gauge *dynamics* (SU(3)'s dynamics is imported — the geometry supplies the container and the count), not the generation mechanism (open, well-posed). Every claim above carries its condition on its face; the tier guide and falsifier register travel with the packet.

The sentence to keep

> Two shapes' worth of information and one measured integer generate the discrete Standard Model — and the same two invariants that build it also forbid the things it must not contain (no GUT, no proton decay, no sterile neutrino): the absences are predictions, and any one of them can kill the theory.